2526-MA378: Class Test ANS (with solutions)

13:00, Wednesday 18 February 2026 **Time Allowed: 40 Minutes**

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Q1. (10 marks) Answer "True" or "False" for each of the following. You do not need to explain your answer.

(a) If $p_k(x)$ is a polynomial of degree k, the $p(x) = \sum_{k=0}^{N} p_k(x)$ is a polynomial of degree $N+1$. ANS False

(b) If $p_{2N+1}(x)$ is the Hermite interpolation polynomial that interpolates a function f at $N+1$ interpolation points, then $p'_{2N+1}(x_i) = f'(x_i)$ at each of those points. ANS True

(c) If $S(x)$ is the natural cubic spline interpolant to any function f at the interpolation points $\{x_0, x_1, \dots, x_N\}$, then $S''(x_i) = 0$ for $i = 0, 1, \dots, N$. ANS False

(d) Let $L_k(x)$ be the kth Lagrange Basis Polynomial for a set of interpolation points $\{x_0, x_1, \dots, x_N\}$, and $\pi_{N+1}(x)$ the associated nodal polynomial, then $L_k(x) = \dfrac{\pi_{N+1}(x)}{x - x_k}$. ANS False

(e) Let $L_k(x)$ be the kth Lagrange Basis Polynomial for a set of interpolation points $\{x_0, x_1, \dots, x_N\}$. Then $L_k(x)$ has exactly $N-1$ zeros in $[x_0, x_N]$. ANS False

Q2. (45 marks)

(a) Let f be a function such that $f(1) = 0$, $f(2) = 1$, $f(3) = 5$ Write down the Lagrange form for the polynomial, $p_2(x)$, that interpolates f at the points $x_0 = 1$, $x_1 = 2$, and $x_2 = 3$.

Answer: [20 MARKS] The Lagrange for an interpolant of degree 2 to f is

$$p_2(x) = L_0(x)f(x_0) + L_1(x)f(x_1) + L_2(x)f(x_2).$$

For this problem

$$\begin{aligned}
L_0(x) &= \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)} = \frac{(x-2)(x-3)}{(-1)(-2)} = \frac{1}{2}(x-2)(x-3),\\
L_1(x) &= \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)} = \frac{(x-1)(x-3)}{(1)(-1)} = -(x-1)(x-3),\\
L_2(x) &= \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)} = \frac{(x-1)(x-2)}{(2)(1)} = \frac{1}{2}(x-1)(x-2).
\end{aligned}$$

Using that $f(x_0) = 0$, $f(x_1) = 1$, and $f(x_2) = 5$, we get that the Lagrange form is

$$p_2(x) = (1)L_1(x) + 5L_2(x) = -(x-1)(x-3) + \frac{5}{2}(x-1)(x-2).$$

If you wish, you can simplify this as $p_2(x) = (3/2)x^2 - (7/2)x + 2$ (Note: It is not important for this test, but $f(x) = x! - 1$. So it is related to the famous gamma function, used in statistics, quantum mechanics, fluid dynamics, number theory, etc. It's formula is a little convoluted so we won't us it here.)

(b) Evaluate $p_2(5/2)$. ANS [10 MARKS] $p_2(5/2) = (3/4) + 5(3/8) = 21/8 = 2.625.$

(c) Suppose we are told that $|f'''(x)| \leq 30.37$. What bound does (1) give for $|f(5/2) - p_2(5/2)|$?

Answer: [15 MARKS] Equation (1) gives $f(5/2) - p_2(5/2) = \dfrac{f'''(\tau)}{3!}\pi_3(5/2)$, for some (unknown) $\tau \in (1,3)$. $\pi_3(x) = (x-1)(x-2)(x-3)$, so $\pi_3(5/2) = -3/8$ So the bound on $|f(5/2) - p_2(5/2)|$ is

$$|f(5/2) - p_2(5/2)| \leq \frac{\max_{1 \leq x \leq 3} |f'''(x)|}{3!} |\pi_3(5/2)| \leq \frac{30.37}{6}\frac{3}{8} = 1.898125. \qquad (*)$$

Q3. (30 marks)

(a) Let l be the piecewise linear interpolant to the problem in Q2(a). Evaluate $l(3/2)$.

Answer: [10 MARKS]

$$l(x) = \begin{cases} f(x_0)\frac{x-x_1}{x_0-x_1} + f(x_1)\frac{x-x_0}{x_1-x_0} & x_0 \le x \le x_1 \\ f(x_1)\frac{x-x_2}{x_1-x_2} + f(x_2)\frac{x-x_1}{x_2-x_1} & x_1 < x \le x_2 \\ 0 & \text{otherwise} \end{cases}$$

To answer this question, we only need the formula for $l(x)$ on $[1,2]$, but for completeness:

$$l(x) = \begin{cases} x-1 & 1 \le x \le 2 \\ -(1)(x-3) + 5(x-2) = 4x-7 & 0 < x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

Then $l(3/2) = 1/2$.

(b) Suppose that $\|f''\|_\infty \le 11.12$. Use (2) to give an upper bound for $\|f(x) - l(x)\|_\infty$.

Answer: [10 MARKS] For this problem, $h = 1$, so from (2) we have

$$\|f - l\|_\infty \le \frac{h^2}{8}\|f''\|_\infty \le \frac{11.12}{8} = 1.39.$$

(c) If one uses $N+1$ equally spaced interpolation points on the interval $[1,3]$, what value of N would you have to choose so that $\|f - l\|_\infty \le 10^{-6}$?

Answer: [5 MARKS] We need h such that $\frac{h^2}{8}\|f''\|_\infty \le 10^{-6}$. That is $h \le \sqrt{8 \times 10^{-6}/11.12} \approx \sqrt{7.194245 \times 10^{-7}} = 8.4812 \times 10^{-4}$. Also, $h = (x_2 - x_0)/N = 2/N$, so $N \ge 2/h = 2357.965$. **Take** $N = 2358$.

Q4. (`10 marks`) Suppose that S is the **natural cubic spline** interpolant for the problem in Q2(a). If it is given that $\|f^{(4)}\|_\infty = 30.37$, use (3) to give an upper bound for $\|f - S\|_\infty$.

Answer: We'll use (3), and that $h = 1$ to get

$$\|f - S\|_\infty \le \frac{5h^4}{384}\|f^{(4)}(x)\|_\infty \le \frac{5}{384}30.37 \approx 0.39644$$

Q5. (`5 marks`) Let f be some continuous non-constant function, and $\{x_0, x_1, \ldots, x_N\}$ be a set of interpolation points. Let l be the piecewise linear interpolant to f at these points, and S the natural cubic spline interpolant. Give *two distinct* scenarios where by $\|l - S\|_\infty = 0$.

Answer: **Scenario 1**: $f(x)$ is a polynomial of degree 1. Therefore, $f(x) = l(x)$. Furthermore, since $f''(x) \equiv 0$, we get that $f(x) = S(x)$. Hence $f(x) = S(x)$.
Scenario 2: For any f, if $N = 1$ then $S(x)$ constitutes a single cubic polynomial with the property that $S(x_0)'' = S''(x_N) = 0$. Thus S is, in fact, linear. And since both $l(x)$ and $S(x)$ interpolate f at two points, they must be equal.

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Some useful formulae.

- Cauchy's theorem: If p_n be the polynomial of degree n that interpolates f at the $n+1$ points $a = x_0 < x_1 < \cdots < x_n = b$. Then, for any $x \in [a,b]$ there is a $\tau \in (a,b)$ such that

$$f(x) - p_n(x) = \frac{f^{(n+1)}(\tau)}{(n+1)!}\pi_{n+1}(x), \tag{1}$$

where $\pi_{n+1}(x) = \prod_{i=0}^{n}(x - x_i)$ denotes the nodal polynomial.

- $\|g\|_\infty$ denotes $\max\limits_{x_0 \le x \le x_N} |g(x)|$.

- If l be the linear spline interpolant to a function f on the equally spaced points $a = x_0 < x_1 \cdots < x_N = b$ with $h = x_i - x_{i-1} = (b-a)/N$, then

$$\|f - l\|_\infty \le \frac{h^2}{8}\|f''\|_\infty, \tag{2}$$

- If S is the Natural Cubic Spline Interpolant to a function f at the equally spaced points $\{x_0 < x_1 < \cdots < x_N\}$ with $(x_N - x_0)/N =: h$, then

$$\|f - S\|_\infty \le \frac{5h^4}{384}\|f^{(4)}(x)\|_\infty \tag{3}$$

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